SUZANNE CORY
High School

Unit 3 Chemistry (2022)

SAC 1

Reading time: 5 minutes
Writing time: 70 minutes

Name: _____ Teacher: _____

QUESTION AND ANSWER BOOK

Section	*Number of marks*
Section A	20
Section B	23
Section C	11
	Total: 54 Marks

- Students are permitted to bring to the assessment: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator.
- Students are **NOT** permitted to bring into the assessment rooms: blank sheets of paper and/or white out liquid/tape.

Materials supplied

- Question and answer book.
- A data book.

Instructions

- Write your **name**, and your **teacher's** name in the space provided above on this page.
- All written responses must be in English.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the examination room.

SECTION A: Fuels (20 marks)

1. Biomass can be used to produce the potentially useful fuels biodiesel, biogas or bioethanol.

   a. It's becoming evident that the world needs to switch to renewable fuels such as biodiesel.

      i. Name a raw material that could be converted to biodiesel. (1 mark)

      _____

      ii. A major component of biodiesel (when made from a particular raw material) is methyl linoleate, $C_{17}H_{31}COOCH_3$. Name X and Y and balance the equation below for the production of methyl linoleate (3 marks)

      Reagent X:_____

      Product Y:_____

      iii. Write a balanced equation for the complete combustion of methyl linoleate (2 marks)

      iv. The heat of combustion for methyl linoleate is 11690 kJ mol^{-1}. Calculate the mass of CO_2 produced when 1.0kJ of energy is released by its complete combustion. (2 marks)

$$\begin{array}{l} CH_2OCOC_{17}H_{31} \\ CHOCOC_{17}H_{31} \\ CH_2OCOC_{17}H_{31} \end{array}$$

      v. Explain why in cold regions petrodiesel may be preferred for use in diesel vehicles rather than biodiesel, referring to relevant chemical principles. (3 marks)

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   b. In Australia 0.5% of our electricity is produced using biogas

i. Identify the major organic component of biogas. (1 mark)

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ii. Write a balanced thermochemical equation for the combustion reaction of your answer to part (i) in the presence of excess oxygen (2 marks)

iii. Briefly explain how biogas is produced (1 mark)

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iv. Biogas is known as a renewable resource, what defines a renewable resource? (1 mark)

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_____

c. Bioethanol (CH_3CH_2OH) can be produced from sugar cane crops

i. State ONE environmental disadvantage of producing ethanol this way. (1 mark)

_____

_____

ii. What mass of octane will produce the same amount of energy that is released when 40 L of propane at 105 kPa at 26°C is completely combusted. (3 marks)

SECTION B: Energy from fuels (23 marks)

1. Methane and methanol will both burn in air. The reactions are described by the equations

$$CH_4(g) + 2O_2(g) - \quad CO_2(g) + 2H_2O(l); \qquad \Delta H = -890 \text{ kJ mol}^{-1}$$

$$2CH_3OH(l) + 3O_2(g) - \quad 2CO_2(g) + 4H_2O(l); \qquad \Delta H = -1452 \text{ kJ mol}^{-1}$$

a. If 2 mole of methane and 2 mole of methanol are completely burnt in separate experiments, which will release the most energy? (1 mark)

b. If each of the above reactions is used to produce 1000 kJ of energy, which one will release the most carbon dioxide? (2 marks)

c. In an exothermic reaction compare how the strength of the bonds in the reactants is related to the strength of the bonds in the products (1 mark)

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2. Carbon, in the form of graphite, burns in oxygen according to the following equation

$$C(s) + O_2(g) \rightarrow CO_2(g) \qquad \Delta H = -394 \text{ kJ mol}^{-1}$$

a. The activation energy for this reaction is high

i. Explain what is meant by the expression 'activation energy' (1 mark)

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_____

_____

ii. Suggest why the activation energy for this reaction is high. (1 mark)

_____

_____

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b. Draw an energy profile for the reaction of graphite with oxygen with the activation energy of 172 kJ mol^{-1}. Label the specific reactants, specific products, activation energy and ΔH on your profile and appropriate scale (2 marks)

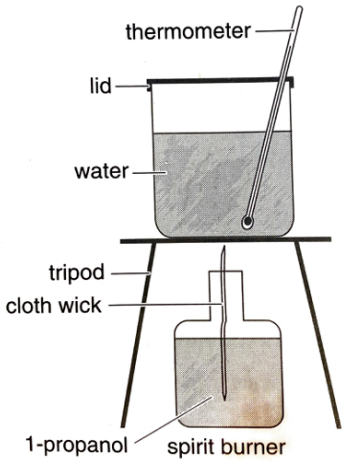
c. On the same energy profile draw a possible catalysed reaction profile (1 mark)

3. A student burns a 4.50g sample of graphite in excess oxygen. The heat evolved is absorbed by 500 g of water.

a. Calculate the amount of heat energy evolved when the graphite burns (in kilojoules) (2 marks)

b. Calculate the expected rise in temperature of the water, given that 4.18 J is required to raise the temperature of 1g of water by 1.0 (1 mark)

4. A student conducted an experiment to investigate the combustion of ethanol. Liquid ethanol was placed in a spirit burner, the wick lit and the heat released used to heat a beaker of water. The apparatus used and results obtained are shown below.



Results: mass of ethanol burned = 1.20 g

mass of water heated = 150 g

rise in temperature of water = 51°C

Observation: it was difficult to start the ethanol burning

a. Calculate the energy (in kJ) absorbed by the water in the beaker (2 marks)

b. Calculate the theoretical energy released by the burning ethanol. (2 marks)

c. Calculate the percentage of energy from the burning ethanol which was actually used to heat the water (1 mark)

d. Suggest one safety concern for this experiment (1 mark):

_____

_____

_____

e. Suggest a way to improve the accuracy of this experiment (2 marks)

5. Gaseous ethanol undergoes complete combustion in the presence of excess oxygen according to the following reaction

$$2C_2H_6O(g) + 7O_2(g) \quad 4CO_2(g) + 6H_2O(l)$$

If 4.2 L of ethanol is burnt in the presence of excess oxygen under SLC conditions, calculate the volume of carbon dioxide produced. (1 marks)

6. Copper forms two oxides, Cu_2O and CuO. The energy change when each oxide is formed is given by the equations below. (2 marks)

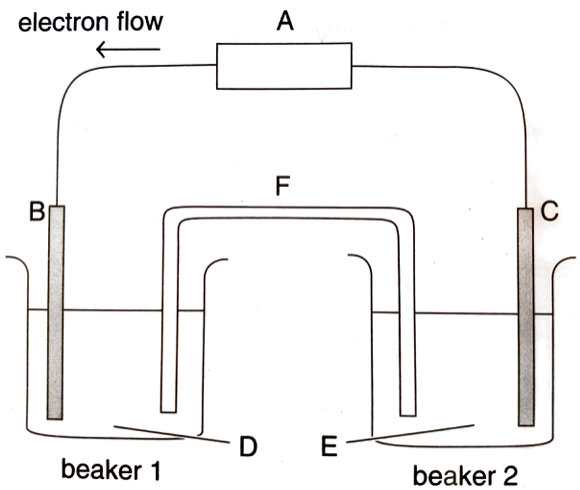
$2Cu(s) + O_2(g) \quad 2CuO(s) \qquad \Delta H = -310.4 \text{ kJ mol}^{-1}$
$4Cu(s) + O_2(g) \quad 2Cu_2O(s) \qquad \Delta H = -338 \text{ kJ mol}^{-1}$

Use the information to calculate the ΔH for the reaction
$Cu_2O(s) \quad Cu(s) + CuO(s)$

SECTION C: REDOX (11 marks)

1. A student is provided with the materials listed below and is told to use whichever materials are needed to make a galvanic cell which will generate and measure a voltage.

Beakers, thermometer, power pack, light globe, insulated copper wire, voltmeter, filter paper, funnel, stirring rods, sand paper, 1.0 M solution of copper(II) nitrate, 1.0 M solution of potassium nitrate, 1.0 M of iron(II) nitrate, deionised water, strips of copper, strips of iron.



a. Write an appropriate aim for this experiment (1 mark):

_____

_____

b. In the table below, identify the following components (4 marks)

Electrode C	
Electrode B	
Electrolyte D	
Ions involved in component F	

c. Write a balanced half-equation for the cathode reaction. (1 mark)

d. Describe two changes you would expect to observe in the beakers if a current flowed for a considerable length of time. (2 marks)

Change 1:

_____

_____

_____

Change 2:

_____

_____

_____

e. Calculate the potential difference of this galvanic cell (1 mark)

f. Would you expect a change to occur in the concentration on NO_3^- (aq) in beaker 2 when the cell has been producing current for a considerable length of time? Explain your response. (2 marks)

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